Venkata Lakshmi Parimala Pasupuleti

Senior AI Engineer · LangGraph · Multi-Agent · RAG · Evaluation Harnesses · Internal AI Platforms · Python · PropTech · 9yr Production

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PROFESSIONAL SUMMARY

Senior AI engineer with 9+ years of production experience and 2+ years building agentic systems, evaluation harnesses, and RAG pipelines - exactly what RealPage's Internal AI Center of Excellence requires. AgentProbe is the direct proof point: a production evaluation platform I designed and shipped end-to-end with a 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency), LLM-as-Judge scoring, pass@k multi-trial harness, CI-integrated regression gate blocking drift before deployment, and multi-provider LLM routing (Anthropic, OpenAI, Ollama) - the custom evaluation harness and reusable AI patterns the CoE is building internally. LeaseI is the multi-agent proof point: an 8-agent LangGraph StateGraph (the exact framework named in this role) with conditional routing, dynamic task delegation, shared long-term memory, tool invocation, and deterministic HITL guardrails for regulated decisions - paired with ChromaDB RAG, semantic search, and cosine reranking. I use Claude Code and AI coding tools daily to accelerate engineering delivery responsibly, and I actively mentor engineers on agentic AI patterns, LLM integration, and production best practices - the enablement responsibility at the core of this role. At Entrata (RealPage's PropTech competitor), I owned production systems for 7 years (Python, AWS, Docker, RabbitMQ, FastAPI) at 99.5% uptime and 100% PHI/HIPAA compliance - the reliability and regulated-data standards enterprise AI requires. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Agentic AI & Evaluation: Python (9yr), LangGraph, LangChain, Multi-Agent Orchestration, Generative AI, Tool Calling, Function Calling, Multi-Step Agentic Workflows, RAG Pipelines, ChromaDB, Vector Databases, Semantic Search, Semantic Retrieval, Enterprise Search, Knowledge Management, Prompt Engineering, Prompt Versioning, Evaluation Frameworks, Evaluation Harnesses, LLM-as-Judge, Custom Evals, Hallucination Detection, Groundedness, CI Regression Gates, HITL Guardrails, MCP, Anthropic API, OpenAI API, Ollama, Claude Code, GitHub Copilot

Infrastructure & Platform: AWS (S3, EC2, RDS, Glue), Cloud-Native Architecture, Docker, Microservices, FastAPI, Flask, REST APIs, API Design, TypeScript, JavaScript, React.js, RabbitMQ (event-driven async, retry, idempotency), Redis, PostgreSQL, MySQL, Oracle, GitHub Actions CI/CD, Playwright, Automated Testing, Observability, Logging, Performance Engineering, Linux (9yr production)

Domain & Enablement: PropTech (7yr Entrata), Privacy & Data Protection, PHI/HIPAA Compliance, Regulated Data Handling, Audit Logging, RBAC, Internal Developer Tools, AI Adoption Enablement, Engineering Productivity, Technical Mentoring (600+ students, 5+ engineers), Workshops & Office Hours, AI Champions Programs, Code Reviews (300+), Agile/Scrum (120+ sprints), Cross-functional Delivery, PyTorch, scikit-learn, Pandas, NumPy

PROJECTS

AgentProbe: Internal AI Evaluation Platform - Custom Eval Harness & LLMOps

Evaluation Harness · LLMOps · CI Gate · RAG · Open Source

Python · FastAPI · LangGraph · ChromaDB · MCP · GitHub Actions · Anthropic API · OpenAI API · Ollama · Claude Code | github.com/venkatalakshmiparimala/AgentProbe

Production internal AI evaluation platform - the reusable evaluation harness, observability engine, and reference implementation pattern the RealPage AI CoE is building at enterprise scale. **Evaluation harness:** 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency); LLM-as-Judge scoring; pass@k multi-trial harness with multi-sampling to surface variance in non-deterministic outputs. **CI regression gate:** GitHub Actions integration blocks quality drift and unsafe output patterns before deployment - making evaluation a first-class engineering practice. **Agentic platform:** multi-provider LLM routing (Anthropic, OpenAI, Ollama); ChromaDB RAG; tool-calling patterns; MCP integration; 15-endpoint FastAPI backend. **Enablement:** open-source, documented, enterprise-deployable - built as a reference implementation for engineering teams. Zero regressions across 1,240+ test cases.

LeaseI: Multi-Agent System - LangGraph Orchestration & RAG

LangGraph · Multi-Agent · HITL · RAG · PropTech Domain

Python · LangGraph (8-agent) · FastAPI · ChromaDB · SQLite · Playwright · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production multi-agent system built with LangGraph (the exact framework named in this role) - demonstrating the agentic workflow and orchestration patterns the CoE will standardize across RealPage engineering teams. **Agent orchestration:** 8-agent LangGraph StateGraph with conditional routing, dynamic task delegation based on confidence scoring, and tool-calling across retrieval, structured data, and configuration agents. **RAG infrastructure:** ChromaDB vector database, section-aware chunking, cosine reranking, multi-source retrieval. **Long-term memory:** shared SQLite knowledge repository persisted across all agents and sessions. **Guardrails:** deterministic HITL escalation for regulated decision paths - safety boundary enforced in code. 93.6% precision, 87.9% F1 across 1,240 test cases. PropTech/affordable housing domain.

AI on FHIR: Enterprise AI Integration - Tool Use, APIs & Compliance

Enterprise APIs · Tool Use · HIPAA · Agentic · Full Stack

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · FHIR REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

HIPAA-compliant agentic system integrating LLM-powered tool use with enterprise REST APIs - the AI-to-system integration pattern the CoE will enable across RealPage engineering teams. LLM translates natural-language queries into structured FHIR API calls; RBAC with authentication and full data isolation; complete audit trail; ~60% reduction in manual retrieval time; Docker containerization. Demonstrates secure, observable AI integration with enterprise systems under compliance constraints.

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Production Agentic AI Delivery

- Designed and deployed a production agentic AI system (LangGraph, FastAPI, Python) for an external customer - end-to-end ownership from requirements through architecture, implementation, evaluation harness, prompt versioning, and production deployment in weeks; built guardrails, validation gates, and HITL review for regulated decision paths; created playbooks and documentation for ongoing operation.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate - AI Enablement

- Mentored 600+ students in Python, SQL, ML, and software engineering; built ML pipelines (PyTorch, scikit-learn, R^2=0.977); designed observability dashboards improving comprehension 40%; delivered complex AI/data concepts to non-technical stakeholders - the technical enablement and communication pattern central to this role.

Entrata - Apartment & Affordable Housing SaaS (RealPage Competitor)

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Internal developer tools and engineering productivity** - built internal debugging tools reducing triage time 40%; Python/Bash/PL-SQL automation eliminating 80% of manual engineering workload; maintained bug count below 10 for 6+ consecutive months; 200+ SQL optimizations (30% performance gain); GitHub Actions CI/CD with automated testing throughout; leveraged Claude Code and AI coding tools to accelerate engineering delivery.
- **Production platform engineering** - Python (9yr), FastAPI/Flask REST APIs, PostgreSQL/MySQL/Oracle, Docker microservices (50% scalability improvement), RabbitMQ event-driven async with retry logic and idempotency, AWS S3/EC2/RDS/Glue, Redis; 100% PHI/HIPAA compliance; zero security incidents 7yr; 99.5% uptime; 20+ production modules across 120+ Agile sprints.
- **Technical mentoring and cross-functional enablement** - mentored 5+ engineers on system design, agentic AI patterns, code quality, and production practices; 300+ peer code reviews; presented solutions to engineering and non-engineering stakeholders; cross-functional delivery with product, compliance, and operations teams across 120+ sprints.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (Python, PostgreSQL) for production PropTech SaaS; data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017